

PROJECT PRAHARI (■■■■■■■■■)

AI-Powered Satellite Collision Assessment & Autonomous Risk Triage
Comprehensive Project Handbook, Literature Review & Viva Cheat Sheet

1. The Core Problem in Space (Why this Project Exists)

The Space Traffic Crisis: Over 36,000 trackable orbital debris fragments (>10 cm) and 10,000+ active satellites orbit Earth in Low Earth Orbit (LEO) at hypervelocity speeds exceeding **7.8 km/s (28,000 km/h)**. At this velocity, even a 1-centimeter paint fleck or screw possesses the kinetic explosive force of a hand grenade.

What is a Conjunction & CDM? Whenever radar/telescope tracking networks (e.g. US Space Surveillance Network / ESA) detect two objects approaching within a safety sphere, they issue a standardized telemetry report called a **CDM (Conjunction Data Message)**. A CDM contains positions, velocities, 3D covariance uncertainty matrices, and estimated miss distances.

The Human Bottleneck: Satellite operators receive **100,000+ CDMs every single day**. Over 99.9% are benign false alarms due to sensor noise. However, human orbital engineers spend days manually running orbit propagators. If a high-risk collision isn't identified 24 to 72 hours before closest approach, there is no time left to calculate and fire satellite avoidance thrusters.

2. What Project Prahari Does

Prahari is an autonomous end-to-end mission-control triage platform. Instead of days of human review, Prahari uses an **optimized 100-feature XGBoost ML engine** that inspects incoming CDMs in milliseconds, classifies the true collision risk, assigns actionable risk bands, and provides 3D visual encounter reconstructions.

Alert Band	Log10 Risk Threshold	Probability	Operational Action Required
CRITICAL	≥ -4.0	≥ 1 in 10,000	Immediate Thruster Maneuver (CAM) mandatory.
HIGH	-5.0 to -4.0	1 in 10k to 1 in 100k	Task radar tracking; compute avoidance burns.
ELEVATED	-6.0 to -5.0	1 in 100k to 1 in 1M	Monitor follow-up CDMs; stand-by status.
LOW	< -6.0	< 1 in 1,000,000	Benign approach; automated archive (no burn).

3. Base Paper, Literature Review & Competition

The Base Benchmark: This work builds upon the **ESA Kelvins Collision Avoidance Challenge** conducted by the European Space Agency. The dataset contains thousands of real and simulated conjunction events across ESA operational satellites (Sentinel, CryoSat, Swarm) and tracking debris.

Method / Baseline	Core Mechanism	Critical Weakness / Flaw	Prahari Advantage
Physics-Only (Foster 2D / Alfriend 3D)	Calculates hard covariance overlap integral analytically.	Breaks down when covariance is non-Gaussian or atmospheric drag fluctuates.	Combines covariance + kinematics + solar flux non-linearly.
Logistic Regression / Linear SOTA	Linear boundary based on miss distance & velocity.	Cannot model non-linear interactions between 3D sigma matrices.	Tree ensembles capture multidimensional feature interactions.
Uncalibrated Default ML Classifiers	Default 0.5 classification threshold on raw data.	High False Negatives (misses 40-50% of real collisions).	Calibrated -6.0 threshold gives 92.7% Recall on critical events.

4. Why XGBoost & How it Works (The Simple Dart Analogy)

What is XGBoost? XGBoost stands for *eXtreme Gradient Boosting*. It is an ensemble of decision trees that learn sequentially from each other's mistakes.

- **The Sequential Correction Process:**
 1. **Tree 1** looks at *Miss Distance* and predicts an initial risk score. Its prediction is slightly off (it has a 'residual error').
 2. **Tree 2** is trained specifically to predict the *error made by Tree 1* using *Covariance Sigma R/T/N*.
 3. **Tree 3** predicts the remaining residual error using *Solar Radio Flux (F10.7)* and *Time-to-TCA*.
 4. The final risk score is the weighted sum of all trees, resulting in ultra-precise classification.
- **Top 5 Predictive Features in Telemetry:**
 1. **Mahalanobis Distance:** Euclidean distance scaled by orbital position uncertainty bubbles (22.4% importance).
 2. **Closest Miss Distance:** Minimum physical separation in meters (18.5% importance).
 3. **Relative Encounter Speed:** Relative encounter velocity in m/s (14.2% importance).
 4. **Time to TCA (Time to Closest Approach):** Days remaining until the encounter (11.8% importance).
 5. **Positional Covariance Determinants (Det C):** Total 3D volume of uncertainty (16.5% combined importance).

5. System Architecture & Tech Stack

Layer	Technologies	Key Role & Responsibility
AI Engine	XGBoost, Scikit-learn, Joblib	Loads 100-feature model; executes sub-2ms risk inference and SHAP importances.
Backend API	FastAPI, Uvicorn, Pandas, Python 3.10+	Asynchronous REST API (<code>/api/events`</code> , <code>/api/predict-cdm`</code> , <code>/api/model-info`</code>).
Frontend UI	React 18, Vite, TypeScript, TailwindCSS	Responsive glassmorphism dashboard, alert counters, interactive search & filter.
3D Graphics	Three.js, React-Three-Fiber, Drei, Leaflet	Full 3D orbital encounter scene with uncertainty ellipsoids and 2D ground track.

6. Top 10 Viva & Interview Questions (Quick Revision Sheet)

Q1: What is TCA and why is it critical?

Answer: TCA stands for **Time of Closest Approach**. It is the exact second when two orbital objects reach their minimum distance. Collision Avoidance Maneuvers (CAM) must be executed at least 24-48 hours before TCA to save fuel and ensure trajectory clearance.

Q2: What is a CDM (Conjunction Data Message)?

Answer: A CDM is an international standard telemetry message (CCSDS standard) published by space surveillance tracking networks containing orbital state vectors, velocities, and 3D uncertainty covariances (σ_r , σ_t , σ_n) for two approaching objects.

Q3: Why do we express collision risk in log10 format (e.g. -4.0)?

Answer: Collision probabilities in space are tiny numbers like 0.0001 (10^{-4}) or 0.000001 (10^{-6}). Expressing them as $\log_{10}(\text{Risk})$ converts these exponential fractions into a clear, linear operational scale ($-4.0 = 10^{-4}$, $-6.0 = 10^{-6}$).

Q4: Why prioritize Recall (92.7%) over Precision?

Answer: In aerospace safety, a **False Negative (missed collision)** is **catastrophic** (satellite destruction and Kessler cascade). A False Positive only causes an engineer to inspect the data. High Recall guarantees almost zero true collision threats slip through.

Q5: Why choose XGBoost over Deep Neural Networks for this task?

Answer: CDMs are tabular telemetry data with 100 heterogeneous numerical features and missing tracking data (NaNs). XGBoost natively handles missing values, trains in seconds, runs inference in < 2ms, and outperforms deep neural networks on tabular datasets.

Q6: What is the Mahalanobis Distance?

Answer: It measures the distance between two objects **scaled by their combined 3D covariance uncertainty bubbles**. A 500m miss distance inside a 2,000m uncertainty bubble is dangerous, whereas a 500m miss with a 10m uncertainty bubble is safe.

Q7: What is the Kessler Syndrome?

Answer: A cascading collision scenario where debris collisions create more debris, triggering a chain reaction that could render Low Earth Orbit unusable for communications, navigation, and science for centuries.

Q8: How does Prahari handle multiple CDMs over time for a single event?

Answer: As ground radars make newer observations closer to TCA, multiple CDMs are received. Prahari tracks the chronological risk evolution and evaluates the latest available telemetry state for accurate decision-making.

Q9: What happens when a user uploads a new custom CDM in the app?

Answer: The frontend sends the CSV to `/api/predict-cdm`. The FastAPI backend feeds the telemetry into the 100-feature XGBoost model and returns the predicted risk, alert band (CRITICAL/HIGH/LOW), and top SHAP feature drivers in < 10ms.

Q10: What makes Prahari a standalone deployment?

Answer: Prahari packages its pre-computed 2,167 event archive and ML model weights locally. Double-clicking `START.bat` checks dependencies and launches both backend and frontend without external cloud databases or internet requirements.